

# Project Performance Audit & Cost Report

Generated on July 09, 2026 | Enterprise Sourcing Live Feed

## Executive Summary

This is a Residential building with 5 structural divisions, including 2 bedrooms, 2 bathrooms, 1 kitchen, and 1 hall. Based on the uploaded blueprint, the estimated construction cost is ₹42,12,988 using Standard quality specifications. The estimated completion time is 4 months.

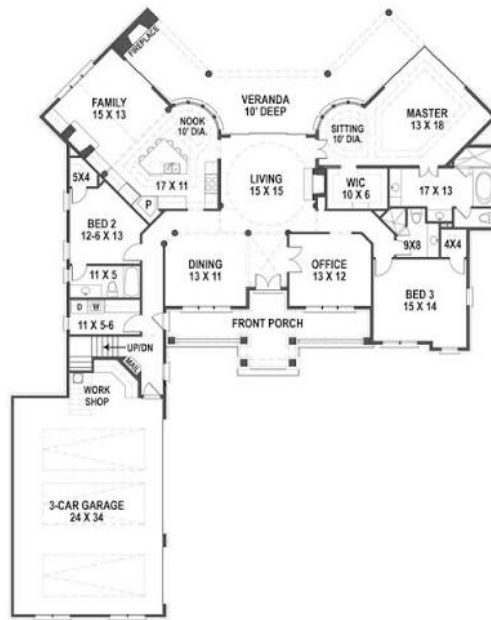
## Project Details & Assigned Personnel

|               |                 |                    |               |
|---------------|-----------------|--------------------|---------------|
| Project Name: | VILLA           | Client Name:       | SITE ENGINEER |
| Location:     | TN              | Building Type:     | Residential   |
| Floors:       | 1               | Material Quality:  | Standard      |
| Assigned PM:  | Marcus Aurelius | Assigned Engineer: | Diana Prince  |

## AI Blueprint Layout Analysis

| Structural Division    | Detected Quantity | Classification          |
|------------------------|-------------------|-------------------------|
| Bedrooms / Cabin Areas | 2 Zones           | Enclosure Space         |
| Bathrooms / Plumbing   | 2 Zones           | Sanitary Block          |
| Kitchen / Cafeteria    | 1 Zone            | Services Block          |
| Hall / Reception Zone  | 1 Zone            | Common Area             |
| Dining / Pantry Area   | 1 Zone            | Amenity space           |
| Balconies              | 1 Zones           | Open Air Extension      |
| Parking Slots          | 1 Zone            | Utility                 |
| Staircase / Lift Core  | 0 Zone            | Vertical Circulation    |
| Doors Identified       | 8 Units           | Apertures / Access      |
| Windows Identified     | 12 Units          | Apertures / Ventilation |
| Total Built-up Area    | 1,200 sqft        | Calculated Area         |

Project Blueprint Layout Drawing Reference:



AI Material Quantity & Sourcing Schedule

| Material Item                       | Estimated Qty | Unit     | Total Cost (INR) |
|-------------------------------------|---------------|----------|------------------|
| Cement                              | 480           | bags     | ₹1,82,400        |
| Steel Rebar                         | 4.8           | tons     | ₹2,97,600        |
| Bricks                              | 26,400        | pcs      | ₹2,11,200        |
| Sand                                | 96            | tons     | ₹1,15,200        |
| Blue Metal Aggregate                | 72            | tons     | ₹1,00,800        |
| Gravel / Soling                     | 48            | tons     | ₹48,000          |
| Ready-Mix Concrete                  | 36            | cum      | ₹1,62,000        |
| Paint                               | 192           | liters   | ₹49,920          |
| Flooring Tiles                      | 1,380         | sqft     | ₹89,700          |
| Electrical Conduit & Wiring         | 1             | lump-sum | ₹1,08,000        |
| Plumbing Pipes & Fixtures           | 1             | lump-sum | ₹96,000          |
| Roofing/Slab Concrete reinforcement | 1             | lump-sum | ₹1,56,000        |
| Doors                               | 8             | units    | ₹48,000          |
| Windows                             | 12            | units    | ₹48,000          |
| Labor Sourcing Requirement          | 1,020         | man-days | ₹5,61,000        |

Financial Summary & Cost Breakdown

| Cost Category           | Estimated Cost (INR) | Cost Category          | Estimated Cost (INR) |
|-------------------------|----------------------|------------------------|----------------------|
| Materials Base Cost     | ₹17,12,820           | Labor Cost             | ₹5,61,000            |
| Equipment & Machinery   | ₹2,40,000            | Electrical Works       | ₹1,08,000            |
| Plumbing & Sewerage     | ₹96,000              | Painting & Coatings    | ₹49,920              |
| Roofing Works           | ₹1,56,000            | Finishing Works        | ₹2,35,620            |
| Government Approvals    | ₹1,50,000            | Miscellaneous Expenses | ₹90,953              |
| Contingency Amount (5%) | ₹1,70,016            | GST (18% applicable)   | ₹6,42,659            |
| Grand Total Estimate    | ₹42,12,988           |                        |                      |

## Construction Timeline & Project Schedule

| Construction Phase / Stage | Estimated Duration (Weeks) |
|----------------------------|----------------------------|
| Site Preparation           | 1                          |
| Foundation                 | 2                          |
| Column Work                | 2                          |
| Wall Construction          | 2                          |
| Roofing                    | 2                          |
| Electrical Work            | 1                          |
| Plumbing                   | 1                          |
| Flooring                   | 2                          |
| Painting                   | 1                          |
| Finishing                  | 2                          |
| Overall Estimated Duration | 4 Months                   |

## AI Construction Risk Scorecard

| Risk category     | Level  | Telemetry Diagnosis                                           | Mitigation Recommendation                                |
|-------------------|--------|---------------------------------------------------------------|----------------------------------------------------------|
| Delay Risk        | Low    | Slight logistical supply delay risks or approval buffer slip. | Maintain a 14-day schedule buffer.                       |
| Budget Risk       | Low    | Market steel price volatility.                                | Procure major materials at project kickoff.              |
| Safety Risk       | Low    | Scaffolding falls and heavy vehicle movements.                | Enforce mandatory hard-hats and harness usage.           |
| Material Sourcing | Medium | Sourcing delays for premium finishes.                         | Source from multiple supplier agreements.                |
| Weather Risk      | Low    | Seasonal monsoon rainfall interruptions.                      | Perform roofing concrete pours before peak rainy season. |

## AI Sourcing & Value Optimization Recommendations

- Suggested Workforce: 10 active workers (skilled & unskilled)
- Suggested Machinery: Concrete Mixer Machine, Slab Vibrator, JCB Excavator
- Cost Optimization: Optimize cut-lists for steel rebar to minimize waste below 5%.
- Construction Sequence: Run electrical conduiting concurrently with brick laying partitions.
- Material Sourcing Improvements: Use fly-ash blended OPC cement for columns foundation slab to lower footprint.
- Safety Program: Conduct safety toolboxes each morning for working-at-height laborers.

Final Conclusion: This cost and quantity report is generated based on standard mechanical layouts and area bounding boxes detected from the building blueprint. The quantities and rates comply with current Central Public Works Department (CPWD) schedule indices in India and are ready for commercial execution review.

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Marcus Aurelius  
Project Manager

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Sarah Administrator  
Auditor Coordinator